

Orthogonal Ray Projection: A Tangent-Space Visual Measurement Model for Robust Visual-Inertial Odometry

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Abstract—The reprojection error in Visual-Inertial Odometry (VIO) suffers from high nonlinearity due to perspective division, which degrades estimator consistency and robustness, particularly under large depth uncertainty. To address this, we propose a novel visual measurement model, the Orthogonal Ray Projection Error (ORPE), which is formulated in the tangent space of the observation ray. By minimizing the orthogonal distance between the estimated landmark and the measurement ray, ORPE decouples the measurement error from the scalar depth, rendering the residual function linear with respect to the feature position. We derive the exact analytical Jacobians and an uncertainty propagation model, integrating ORPE into both the MSCKF-based OpenVINS and the optimization-based ORB-SLAM3 frameworks. Simulations confirm that ORPE achieves geometric linearity for features, while significantly reducing the system nonlinearity with respect to camera pose. Extensive real-world experiments demonstrate that the proposed method significantly improves trajectory accuracy and estimator consistency in challenging weak-parallax scenarios, while maintaining computational efficiency comparable to standard approaches.

Index Terms—Visual-inertial odometry, orthogonal ray projection, tangent space parameterization, estimator consistency, geometry linearity.

I. INTRODUCTION

VISUAL-INERTIAL Odometry (VIO) and Simultaneous Localization and Mapping (SLAM) constitute the foundational perception layer for autonomous mobile robots [1], [2], [3], uncrewed aerial vehicles (UAVs) [4], [5], and augmented reality (AR) devices [6], [7]. By tightly fusing high-frequency inertial measurements with informative visual constraints, VIO systems achieve robust state estimation even in challenging scenarios characterized by rapid motion, texture scarcity, and temporary occlusions. Over the past decade, the field has evolved from loosely-coupled prototypes to principled probabilistic estimators, which are broadly categorized into filter-based approaches (e.g., MSCKF variants [8], [9]) and optimization-based frameworks (e.g., VINS-Mono [10], ORB-SLAM3 [11]).

From the perspective of data processing, VIO approaches are typically divided into direct and feature-based methods. Direct methods [12], [13] minimize photometric errors directly on pixel intensities, offering high precision in low-texture environments but suffering from sensitivity to illumination changes. Conversely, feature-based methods [14], [15], [16] rely on geometric reprojection errors of extracted landmarks. Owing to their inherent resilience to illumination changes and efficient geometric representation, feature-based approaches demonstrate superior robustness across diverse platforms, making them the dominant paradigm for long-term navigation in complex environments.

However, the standard reprojection error used in feature-based methods suffers from inherent geometric limitations. The projection function involves perspective division, which introduces severe nonlinearity into the measurement model. This nonlinearity manifests as an uneven cost landscape, making the estimator sensitive to linearization errors. Furthermore, for distant features or those with weak parallax, the depth uncertainty creates elongated, non-Gaussian uncertainty ellipsoids. When projected onto the 2D image plane, these uncertainties are distorted, leading to ill-conditioned Jacobians and inconsistent observability. This strong coupling between pose and depth often triggers scale drift and creates numerical instability in both Extended Kalman Filter (EKF) and nonlinear optimizers.

To address this, we propose the Orthogonal Ray Projection Error (ORPE), a novel formulation defined in the tangent space of the observation ray. By minimizing the orthogonal distance between the estimated landmark and the measurement

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To facilitate reproducibility and further research, we release our source code implemented in both OpenVINS and ORB-SLAM3 frameworks at our public repository <https://github.com/BISC-Nav/ORPE>.

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ray, ORPE avoids perspective division and renders the residual function linear with respect to the feature position. Although the system inherently remains nonlinear due to camera orientation, this decoupling significantly expands the convergence basin. We integrate ORPE into both the filter-based OpenVINS and optimization-based ORB-SLAM3 frameworks, supported by rigorously derived analytical Jacobians. Extensive evaluations confirm that ORPE enhances trajectory accuracy and estimator consistency while maintaining comparable computational efficiency comparable to standard approaches.

In summary, the main contributions are as follows:

- 1) We propose the ORPE, a novel visual measurement model defined in the tangent space of the observation ray. This formulation fundamentally eliminates perspective division, resulting in a residual function that is linear with respect to feature position, thereby enhancing numerical stability and handling infinite depth naturally.
- 2) We provide a rigorous mathematical derivation of the uncertainty propagation from image-plane pixel noise to the tangent space covariance, along with exact analytical Jacobians with respect to both IMU poses and 3D features. This theoretical foundation allows for the correct weighting of visual residuals in probabilistic estimators.
- 3) We implement and evaluate ORPE within two distinct VIO paradigms: OpenVINS and ORB-SLAM3. To facilitate reproducibility and future research, we release the complete source code to the community.

The rest of this letter is organized as follows: Section II reviews related work. Section III and IV detail the system integration and ORPE methodology, respectively. Section V provides experimental validation, followed by conclusions in Section VI.

II. RELATED WORK

A. State Parameterization

To address the nonlinearity arising from distant features, Inverse Depth Parameterization (IDP) [17] parameterizes features using the inverse of depth, allowing for the initialization of points at infinity and a better Gaussian approximation. Variants such as Inverse Depth Bundles [18] and anchored representations [19] further optimize computational efficiency by sharing anchors. However, while these methods improve the numerical properties of the state vector, they fundamentally rely on the standard planar reprojection residual. Consequently, the intrinsic geometric limitation remains unresolved within the measurement model; the estimator remains vulnerable to linearization errors when projecting 3D uncertainties onto the 2D plane, particularly for long-range features where depth unobservability degrades the Jacobian conditioning.

B. Geometric Error Models

To circumvent the intrinsic limitations of planar projection that state parameterizations fail to resolve, recent research focuses on reformulating the residual definition directly. One direction eliminates explicit feature depth using epipolar geometry, such as the Normal Epipolar Constraint (NEC) [21], [22]. While these depth-free models reduce dimensionality, they are inherently susceptible to geometric degeneracy under specific motions and lack the capability to explicitly constrain 3D map structures.

Another direction employs ray-based formulations for specific tasks like dynamic tracking [23] or shape representation [24]. However, these approaches typically treat the geometric error as a scalar magnitude. Unlike our proposed ORPE, such scalar formulations discard directional information in the tangent space, lacking the rigorous linearization required to maximize the convergence basin for general state estimation.

III. SYSTEM OVERVIEW

To demonstrate the generality and effectiveness of the proposed method, we integrate ORPE into two representative visual-inertial odometry frameworks: the MSCKF-based OpenVINS [20] and the optimization-based ORB-SLAM3 [11].

A. ORPE-Based MSCKF Framework

OpenVINS follows the standard Multi-State Constraint Kalman Filter (MSCKF) pipeline. The system propagates the state covariance using high-frequency IMU measurements. When a visual feature is lost or marginalized, it triggers the measurement update step. In our implementation, denoted as OpenVINS (ORPE), we modify the MSCKF Update module. Instead of projecting the estimated 3D feature onto the image plane to compute the reprojection error, we project the feature onto the orthogonal basis ($\mathbf{b}_1, \mathbf{b}_2$) constructed from the measurement ray $\mathbf{u}$. As derived in Section IV.A, the residual $\mathbf{r}_{orpe}$ and the corresponding Jacobian matrix $\mathbf{H}$ are computed in the tangent space. These are then utilized to update the sliding window of camera states and the IMU state via standard EKF update logic, effectively reducing the linearization error during the filtering process.

B. ORPE-Based Optimization Framework

ORB-SLAM3 is a state-of-the-art SLAM system based on nonlinear optimization. It maintains a global map and estimates the optimal trajectory by minimizing a cost function represented by a factor graph. In the modified ORB-SLAM3 (ORPE), we introduce a novel factor, termed the Orthogonal Ray Projection Factor, to replace the reprojection edge. This factor connects a MapPoint vertex and a KeyFrame Pose vertex. During the Local Bundle Adjustment, the optimizer minimizes the orthogonal distance between the estimated landmark and the observation ray. Since the proposed residual is linear with respect to the feature position, it provides a broader convergence basin for the Levenberg-Marquardt solver, thereby improving the stability of the optimization, especially for newly initialized or distant landmarks.

IV. METHODOLOGY

A. Orthogonal Ray Projection Model

In this section, we present the mathematical formulation of the proposed ORPE formulation. We first derive the tangent space parameterization and the residual definition. Subsequently, we provide the analytical derivation of the exact Jacobian matrix and discuss the uncertainty propagation from the image plane to the tangent space.

Standard visual measurement models typically project an estimated 3D landmark onto the image plane, minimizing the reprojection error in 2D coordinates. However, this approach introduces the perspective division operation ($x/z, y/z$), which

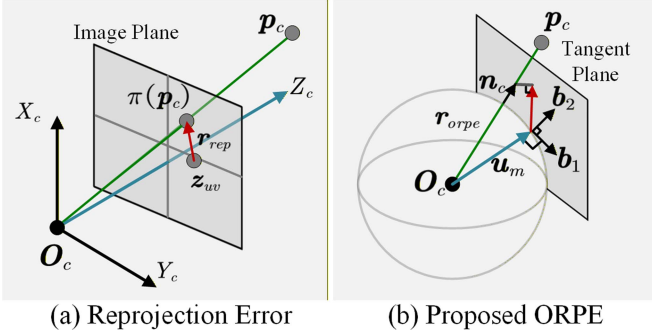


Fig. 1. Geometric comparison of visual error models. (a) Standard Reprojection Error: The 3D feature p_c is projected onto the image plane. The residual r_{rep} corresponds to the Euclidean distance between the measurement z_{uv} and the projection $\pi(p_c)$, which is inherently dependent on the feature depth via perspective division. (b) Proposed Orthogonal Ray Projection Error (ORPE): A tangent plane is constructed at the tip of the observation ray u_m , spanned by the orthonormal basis $B = [b_1, b_2]$ (defined in (2)). The estimated feature p_c (derived in (3)) is projected onto this plane. The residual r_{orpe} , as formulated in (4), is defined as the orthogonal projection of the feature vector p_c onto the basis (b_1, b_2) . This formulation linearizes the measurement model with respect to the feature position.

induces high nonlinearity with respect to the depth. To mitigate this, we propose to formulate the visual residual in the tangent space of the observation ray.

Let $z_{uv} = [u, v]^T$ denote the raw pixel measurement of a feature. Applying the intrinsic camera model $\pi^{-1}(\cdot)$, we obtain the normalized observation vector $u_m \in S^2$ on the unit sphere:

$$u_m = \frac{\pi^{-1}(z_{uv})}{\|\pi^{-1}(z_{uv})\|}, \|u_m\| = 1. \quad (1)$$

This inverse projection operation does not amplify sensitivity to intrinsic calibration, as the geometric error propagation from intrinsic parameters is theoretically equivalent to that of the standard projection model.

To define the error metric, we construct a local tangent plane at u_m . We compute an orthonormal basis $B = [b_1, b_2] \in \mathbb{R}^{3 \times 2}$ that spans the null-space of u_m , satisfying:

$$B^T u_m = 0_{2 \times 1}, B^T B = I_{2 \times 2}. \quad (2)$$

In our implementation, to ensure numerical stability, b_1 is generated by applying the Gram-Schmidt process to u_m and a canonical axis e_{ref} . To handle the singularity when u_m is parallel to the reference axis, we employ a dynamic selection strategy: we set $e_{ref} = [1, 0, 0]^T$ if u_m is nearly parallel to the Z-axis (i.e., $|u_{m,z}| > 0.99$); otherwise, we default to $e_{ref} = [0, 0, 1]^T$. The second basis vector is then determined by $b_2 = u_m \times b_1$. A geometric illustration comparing the proposed tangent space parameterization with the standard planar projection is presented in Fig. 1. As shown in Fig. 1(b), the constructed basis vectors (b_1, b_2) span the plane orthogonal to the observation ray, providing a manifold-aligned error metric.

We define the system state involving the IMU body pose in the world frame, denoted by rotation $R_{wb} \in SO(3)$ and position $p_{wb} \in \mathbb{R}^3$. The camera-to-IMU extrinsic parameters are denoted by (R_{bc}, p_{bc}) . Given a 3D landmark $p_w \in \mathbb{R}^3$, its position in the local camera frame, p_c , is derived through the kinematic chain:

$$p_c = R_{bc}^T (R_{wb}^T (p_w - p_{wb}) - p_{bc}). \quad (3)$$

We define the proposed ORPE residual $r_{orpe} \in \mathbb{R}^2$ as the projection of the estimated feature direction onto the orthogonal tangent basis B . To decouple the residual magnitude from the feature depth, we normalize the estimated point to the unit sphere:

$$r_{orpe}(p_c) = B^T \frac{p_c}{\|p_c\|}. \quad (4)$$

Geometrically, this residual represents the projection of the angular error onto the tangent plane, providing a metric that is invariant to the scale depth. And, explicit chirality checks are enforced throughout the estimation process to ensure that all landmarks remain physically valid.

To linearize the residual for the EKF or optimization solver, we apply the chain rule to derive the Jacobian with respect to the state vector. This is decomposed into the geometric projection Jacobian and the kinematic Jacobian.

First, we derive the derivative of the residual w.r.t. the vector p_c . Let $n_c = p_c / \|p_c\|$ denote the normalized direction vector. We first compute the Jacobian of the normalization operation:

$$\frac{\partial n_c}{\partial p_c} = \frac{1}{\|p_c\|} I - \frac{p_c p_c^T}{\|p_c\|^3} = \frac{1}{\|p_c\|} (I - n_c n_c^T). \quad (5)$$

Applying the chain rule to the residual definition $r_{orpe} = B^T n_c$, we obtain:

$$J_{geo} = B^T \frac{\partial n_c}{\partial p_c} = \frac{1}{\|p_c\|} (B^T - B^T n_c n_c^T). \quad (6)$$

By substituting $r_{orpe} = B^T n_c$ into the equation, we arrive at the compact form of the exact geometric Jacobian:

$$J_{geo} = \frac{\partial r_{orpe}}{\partial p_c} = \frac{1}{\|p_c\|} (B^T - r_{orpe} n_c^T). \quad (7)$$

We explicitly retain $r_{orpe} n_c^T$ to ensure an exact Jacobian satisfying $J_{geo} p_c = 0$. This enforces strict depth invariance, preventing spurious gradients along the depth direction that would otherwise degrade convergence.

We now project J_{geo} onto the state variables using the perturbation model. Let $[\cdot]_{\times}$ denote the skew-symmetric matrix for cross products. The Jacobians with respect to the standard VIO states are derived as follows:

$$H_{p_w} = \frac{\partial r_{orpe}}{\partial p_c} \frac{\partial p_c}{\partial p_w} = J_{geo} R_{bc}^T R_{wb}^T, \quad (8)$$

$$H_{p_{wb}} = \frac{\partial r_{orpe}}{\partial p_c} \frac{\partial p_c}{\partial p_{wb}} = J_{geo} (-R_{bc}^T R_{wb}^T), \quad (9)$$

$$H_{\theta_{wb}} = \frac{\partial r_{orpe}}{\partial p_c} \frac{\partial p_c}{\partial \delta \theta_{wb}} = J_{geo} R_{bc}^T [p_b]_{\times}. \quad (10)$$

where $\delta \theta_{wb}$ represents the rotation error-state of the IMU body frame relative to the world frame, and p_b denotes the 3D feature position expressed in the IMU body frame.

The measurement noise originates from the pixel grid, denoted by $\Sigma_{pix} = \sigma^2 I_2$ [25]. To properly weight the residual in the tangent space, we propagate this covariance using the camera focal length f . Under the small-angle approximation, the covariance for ORPE, Σ_{orpe} , is given by:

$$\Sigma_{orpe} \approx \frac{1}{f^2} \Sigma_{pix}. \quad (11)$$

The propagated covariance is utilized to construct the measurement noise matrix $\mathbf{R}$ in the MSCKF update and the information matrix $\mathbf{\Omega}$ in the factor graph optimization. It is important to note that the scalar approximation in (11) assumes a pinhole projection. For high-distortion lenses, angular resolution varies across the field. To ensure the validity of this assumption, we apply undistortion to project features onto a virtual pinhole plane. While a theoretically exact covariance could be derived, the scalar approximation proves sufficient and numerically stable for the undistorted features.

B. MSCKF Integration With ORPE

In the OpenVINS framework, the MSCKF update is triggered when a feature track is finalized. At this stage, we utilize the proposed ORPE model to construct a geometric constraint on the sliding window states without including the 3D feature position in the state vector.

For a feature $\mathbf{f}_j$ observed by M_j cameras, we stack the individual ORPE residuals $\mathbf{r}_j$ and block Jacobians derived in Section IV.A to form the linearized error equation:

$$\mathbf{r}_j \approx \mathbf{H}_{\mathbf{x},j} \delta \mathbf{x} + \mathbf{H}_{\mathbf{f},j} \delta \mathbf{p}_{w,j} + \mathbf{n}_j, \quad (12)$$

where $\mathbf{H}_{\mathbf{x},j}$ and $\mathbf{H}_{\mathbf{f},j}$ are the stacked Jacobians w.r.t the IMU pose and feature position, respectively. The noise $\mathbf{n}_j$ follows $\mathcal{N}(0, \mathbf{R}_j)$, where $\mathbf{R}_j$ is composed of the tangent space covariances Σ_{orpe} derived in (11).

To eliminate the dependency on the feature error $\delta \mathbf{p}_{w,j}$, we employ the null-space projection. We compute the basis $\mathbf{Q}_2$ for the left null-space of $\mathbf{H}_{\mathbf{f},j}$ (such that $\mathbf{Q}_2^T \mathbf{H}_{\mathbf{f},j} = 0$) and project the residual system:

$$\mathbf{r}_{proj,j} = \mathbf{Q}_2^T \mathbf{r}_j \approx (\mathbf{Q}_2^T \mathbf{H}_{\mathbf{x},j}) \delta \mathbf{x} + \mathbf{n}_{proj}. \quad (13)$$

The projected residual $\mathbf{r}_{proj,j}$ depends solely on the IMU pose error $\delta \mathbf{x}$. This allows us to update the sliding window and IMU states via the standard EKF equations using the projected noise covariance $\mathbf{R}_{proj} = \mathbf{Q}_2^T \mathbf{R}_j \mathbf{Q}_2$.

C. Graph-Optimization Integration With ORPE

In the ORB-SLAM3 framework, we introduce a novel binary edge, the Orthogonal Ray Projection Factor, to replace the standard reprojection constraint in the BA modules.

Let $\mathcal{K}$ be the set of KeyFrames and $\mathcal{P}$ be the set of MapPoints. The optimization seeks to estimate the optimal states $\mathbf{x}^*$ by minimizing the weighted squared errors over the factor graph:

$$\mathbf{x}^* = \arg \min_{\mathbf{x}} \sum_{i \in \mathcal{K}} \sum_{j \in \mathcal{P}_i} \rho(\mathbf{r}_{ij}^T \mathbf{\Omega}_{ij} \mathbf{r}_{ij}). \quad (14)$$

where $\mathcal{P}_i$ denotes the subset of MapPoints observed by KeyFrame i . Here, the residual $\mathbf{r}_{ij} = \mathbf{r}_{orpe}(\mathbf{T}_{wi}, \mathbf{p}_{wi})$ is the geometric error defined in Section IV.A. The information matrix is set as $\mathbf{\Omega}_{ij} \approx f^2 \Sigma_{pix}^{-1}$, which normalized the error based on the pixel noise model. $\rho(\cdot)$ denotes the Huber robust kernel.

The optimization problem is solved using the Levenberg-Marquardt algorithm. At each iteration, the residual function is linearized via first-order Taylor expansion around the current linearization point $\mathbf{x}$:

$$\mathbf{r}_{ij}(\mathbf{x} + \delta \mathbf{x}) \approx \mathbf{r}_{ij}(\mathbf{x}) + \mathbf{J}_{ij} \delta \mathbf{x}, \quad (15)$$

where $\mathbf{J}_{ij}$ is the Jacobian matrix constructed using the exact analytical blocks derived in Section IV.A. Crucially, since the

ORPE residual is linear with respect to the feature position $\mathbf{p}_{w,j}$, this linear approximation is exact for the feature block. This property grants the solver a wider convergence basin, as evidenced in our experimental results.

V. EXPERIMENTAL RESULTS

A. Experimental Setup

We integrated the proposed ORPE into two representative visual-inertial frameworks: the MSCKF-based OpenVINS [20] and the optimization-based ORB-SLAM3 [11]. For clarity, the modified systems are denoted as OpenVINS (ORPE) and ORB-SLAM3 (ORPE), respectively. To comprehensively evaluate the performance across varying environments, we conducted experiments on two standard benchmarks: the EuRoC MAV Dataset [26] (representing indoor, room-scale scenarios) and the RPNG OpenVINS Dataset [20] (representing large-scale outdoor scenarios). Our proposed method was benchmarked against the original OpenVINS and ORB-SLAM3 baselines, which utilize reprojection error model.

The experimental evaluation is structured as follows. We began with a controlled simulation study to explicitly verify the theoretical properties of ORPE. Following this validation, we assessed the system performance on real-world datasets. Given the underlying theoretical connection between filtering and graph optimization, we leveraged the MSCKF framework to conduct an in-depth analysis of the estimator, specifically investigating the advantages of ORPE in terms of consistency, numerical stability, and computational efficiency. To ensure a fair comparison, all experiments were executed on an Intel Core i9-14900HX machine with 32 GB RAM, running Ubuntu 20.04. The intrinsics and extrinsic parameters for the inertial and camera sensors were strictly unified across all competing algorithms to eliminate calibration-induced discrepancies.

B. Simulation Analysis of Residual Linearity

Before evaluating the system performance on real-world datasets, we first conducted a controlled simulation experiment to validate the theoretical properties of the proposed ORPE. Specifically, we analyzed the linearity of the residual function, as high nonlinearity is the primary cause of filter inconsistency and optimization divergence.

1) *Simulation Setup*: We simulated a camera observing a single static 3D feature point. The feature depth Z varied from 0.01 m to 100 m to explicitly cover the ultra-short range where standard VIO measurement models typically degrade due to linearization errors. We introduced random Gaussian perturbations to the linearization point (camera pose $\delta \mathbf{x}$ with $\sigma_x = 0.1$ and feature position $\delta \mathbf{p}$ with $\sigma_p = 0.1$) to simulate challenging estimation errors, and computed the Relative Linearization Error (RLE). The RLE quantifies the discrepancy between the actual residual change and the first-order approximation:

$$RLE = \frac{\|\mathbf{r}(\mathbf{x} + \delta \mathbf{x}) - (\mathbf{r}(\mathbf{x}) + \mathbf{J} \delta \mathbf{x})\|}{\|\mathbf{r}(\mathbf{x} + \delta \mathbf{x}) - \mathbf{r}(\mathbf{x})\|}. \quad (16)$$

2) *Sensitivity Analysis*: As shown in Fig. 2(a), when the camera pose is perturbed, the proposed ORPE consistently exhibits lower linearization error, particularly in the ultra-short range (< 1 m). While the baseline error explodes due to the singularity of perspective division, ORPE maintains numerical stability,

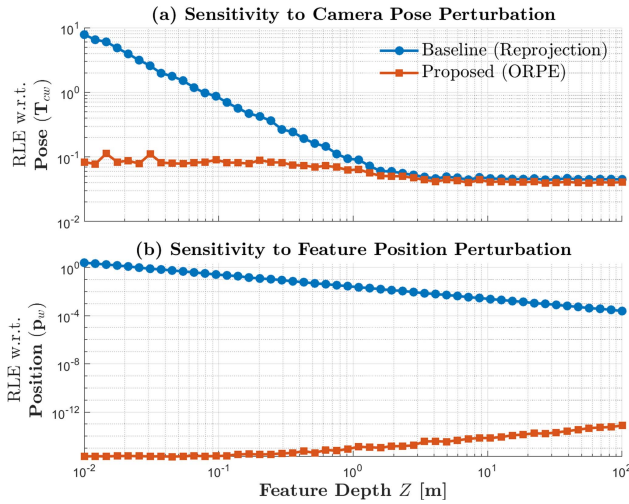


Fig. 2. Comparison of Relative Linearization Error (RLE) with respect to feature depth. (a) RLE under camera pose perturbation. The baseline error explodes in the ultra-short range ($< 1m$) due to the singularity of perspective division, while ORPE remains stable. (b) RLE under feature position perturbation. The baseline reprojection error exhibits high nonlinearity due to perspective division. In contrast, the ORPE maintains an RLE close to machine precision, confirming that the residual is linear with respect to the feature variable.

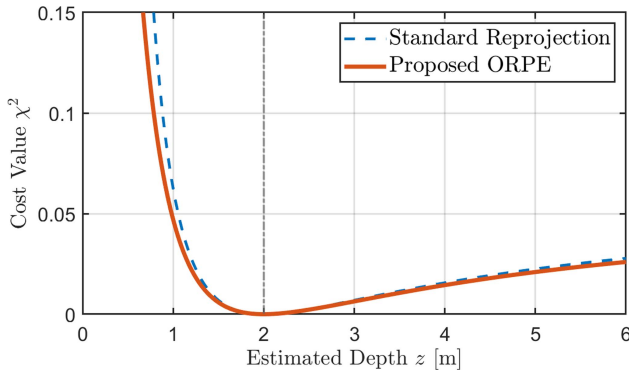


Fig. 3. Comparison of the cost function landscape w.r.t. estimated depth. The standard reprojection (Blue) exhibits a singularity near zero, whereas ORPE (Red) presents a smooth, wide convex basin, demonstrating superior convergence properties.

indicating a wider radius of convergence. A fundamental distinction is observed when the feature position is perturbed, as shown in Fig. 2(b). The baseline RLE increases sharply as depth approached zero. In sharp contrast, the RLE of the proposed ORPE remains at the level of machine precision across the entire depth range. This result confirms that ORPE inherently decouples the measurement from depth. By eliminating the near-field singularity, it ensures exact Jacobians and robust estimation for both extremely close obstacles and distant landmarks.

3) *Convergence Landscape Analysis*: To explicitly demonstrate optimization robustness, Fig. 3 visualizes the cost function landscape with respect to the estimated depth. The standard reprojection error exhibits a sharp singularity as depth approaches zero, creating a narrow, steep valley that hinders convergence from poor initializations. Conversely, ORPE maintains a smooth, quasi-convex profile with a significantly wider convergence basin. This confirms that our formulation inherently

TABLE I
QUANTITATIVE COMPARISON OF ATE RMSE ON THE EUROC DATASET. THE RMSE OF ATE IS REPORTED IN METERS. “ORPE” DENOTES OUR PROPOSED METHOD

Sequence	OpenVINS	OpenVINS (ORPE)	ORB-SLAM3	ORB-SLAM3 (ORPE)
MH01	0.135	0.089	0.075	0.058
MH02	0.260	0.181	0.095	0.066
MH03	0.130	0.111	0.058	0.056
MH04	1.881	0.840	0.168	0.087
MH05	0.566	0.433	0.205	0.062
V101	0.089	0.056	0.037	0.038
V102	0.118	0.056	0.017	0.017
V103	0.078	0.062	0.046	0.030
V201	0.202	0.093	0.063	0.061
V202	0.079	0.064	0.020	0.020
V203	0.170	0.118	0.023	0.016

eliminates the near-field singularity, ensuring reliable convergence even when the initial depth estimate deviates significantly from the true structure.

C. Evaluation of Trajectory Estimation Accuracy

To quantitatively assess the localization accuracy, we employed the Root Mean Square Error (RMSE) of the Absolute Trajectory Error (ATE) as the primary metric. We first evaluated the performance on the EuRoC MAV dataset, which consists of 11 sequences with varying textures and dynamics.

Table I details the ATE RMSE results across all sequences. The proposed ORPE-based methods consistently outperform their respective baselines. Specifically, OpenVINS (ORPE) achieves a significant reduction in error compared to the original OpenVINS, particularly in challenging sequences such as *MH04* and *MH05*. It is important to note that both the baseline and the proposed method utilize the standard IDP. The substantial performance gap observed here is primarily attributed to ORPE’s tangent-space formulation. Unlike the standard reprojection model, ORPE mitigates the high nonlinearity inherent in perspective division. This geometric property ensures more consistent linearization and stable convergence, which is critical for handling the aggressive motion and large uncertainties observed in this sequence. Furthermore, ORB-SLAM3 (ORPE) demonstrates superior accuracy, achieving the lowest RMSE in 9 out of 11 sequences.

We further extended the evaluation to large-scale outdoor environments using the RPNG OpenVINS dataset. These sequences (*neighbor_01* and *neighbor_02*) involve long trajectories where linearization error and drift typically accumulate more severely. As shown in Table II, the proposed method demonstrates even more substantial improvements in the large-scale outdoor RPNG dataset compared to the indoor EuRoC dataset. For instance, in the *neighbor_02*, OpenVINS (ORPE) reduces the RMSE by approximately 57% compared to the baseline, while ORB-SLAM3 (ORPE) reduces it by 50%. The amplified gain in large-scale scenarios validates our depth invariance claim. Unlike standard reprojection, where the $1/z$ scaling biases optimization toward nearby features, ORPE decouples gradient magnitude from depth. This ensures consistent weighting for both near and distant landmarks, effectively suppressing long-term drift in large-scale navigation.

The qualitative improvement is visualized in Fig. 4, which plots the estimated trajectories against the ground truth. As

TABLE II

TRAJECTORY ESTIMATION ERROR ANALYSIS ON THE LARGE-SCALE OUTDOOR RPNG DATASET. WE COMPARE THE MEAN AND RMSE OF THE ATE (IN METERS) FOR THE BASELINE AND ORPE-BASED SYSTEMS. THE PROPOSED METHOD SIGNIFICANTLY REDUCES THE ACCUMULATED DRIFT IN THESE LONG-DURATION ENVIRONMENTS

Sequence	OpenVINS		OpenVINS (ORPE)		ORB-SLAM3		ORB-SLAM3 (ORPE)	
	Mean	RMSE	Mean	RMSE	Mean	RMSE	Mean	RMSE
<i>neighbor_01</i>	15.665	17.325	10.485	12.047	30.635	34.796	19.937	24.402
<i>neighbor_02</i>	31.529	36.202	14.082	15.401	22.403	26.650	11.038	13.282

TABLE III

RUNTIME ANALYSIS OF KEY MODULES (MEAN / STD; UNIT: MS)

Module	Baseline	Proposed (ORPE)
System A: OpenVINS		
Tracking	2.491 / 0.744	2.256 / 0.711
MSCKF Update	0.483 / 0.596	0.558 / 0.765
SLAM Update	5.192 / 1.908	5.438 / 1.702
System B: ORB-SLAM3		
Tracking	22.587 / 5.724	22.995 / 5.894
Local BA	114.480 / 21.086	101.990 / 17.835
Local Mapping	243.003 / 40.885	234.935 / 41.234

observed in the zoomed-in regions, the baseline methods tend to drift significantly at corners and over long durations. In contrast, the ORPE-based trajectories maintain better alignment with the ground truth, confirming that our method effectively mitigates long-term drift in large-scale visual-inertial odometry.

D. Analysis of Estimator Consistency and Depth Invariance

While trajectory accuracy reflects the global performance, it does not fully reveal the underlying properties of the estimator. To investigate the theoretical advantages of the proposed ORPE—specifically its ability to mitigate linearization error and handle depth uncertainty—we conducted a detailed analysis using the MSCKF-based OpenVINS framework. We utilized the *neighbor_01* and *neighbor_02* sequences from the RPNG dataset for this evaluation, as these large-scale outdoor environments contain a mixture of near and far features, which is ideal for testing the system’s robustness to depth variations.

1) *Estimator Consistency Analysis:* The Normalized Estimation Error Squared (NEES) serves as a standard metric to evaluate filter consistency. Fig. 5 depicts the position NEES evolution over time. First, it is important to note that both the baseline and the proposed OpenVINS-ORPE exhibit similar temporal trends. For instance, both systems show high NEES values during the initialization phase and fluctuate synchronously during the subsequent trajectory. This behavior is expected, as it reflects the intrinsic observability constraints and convergence characteristics inherent to monocular VIO systems, regardless of the residual model used. However, a clear distinction lies in the magnitude of the inconsistency. The baseline OpenVINS consistently operates far above the theoretical upper bounds, indicating severe over-confidence. Quantitatively, on the *neighbor_01* sequence, the baseline NEES falls within the 95% confidence interval for only 76.37% of the time steps, whereas the proposed method improves this metric to 83.84%

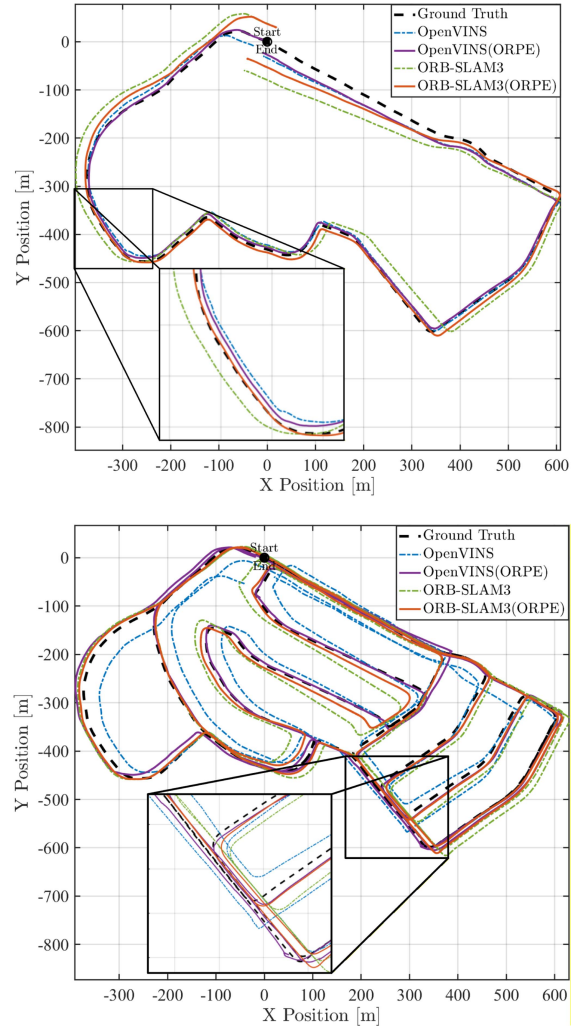


Fig. 4. Qualitative trajectory comparison on the RPNG OpenVINS dataset sequences *neighbor_01* (Up) and *neighbor_02* (Bottom). The estimated trajectories are aligned with the ground truth. The zoomed-in regions highlight that the proposed ORPE methods effectively mitigate the scale and position drift observed in the baseline approaches, particularly during turning and long-distance navigation.

. The improvement is even more pronounced on the challenging *neighbor_02* sequence, where the baseline consistency rate drops to 51.52% , while ORPE maintains a significantly higher rate of 66.01% . This statistical evidence demonstrates that while ORPE does not alter the fundamental observability properties, it effectively reduces the inflation of estimation errors by providing a more numerically stable linearization on the tangent space.

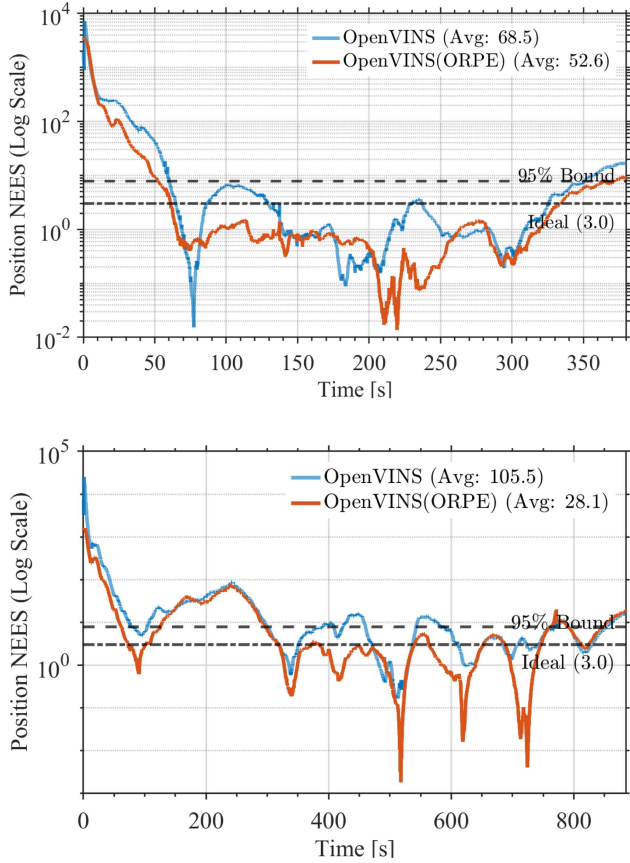


Fig. 5. Position NEES (Log Scale) comparison on the RPNG *neighbor_01* (Up) and *neighbor_02* (Bottom) sequences. The dashed lines represent the ideal consistency value (3.0) and the 95% upper bound. While sharing similar trends, ORPE significantly mitigates the baseline's over-confidence. Quantitatively, the proposed method achieves a relative consistency improvement of 9.8% on *neighbor_01* and 28.1% on *neighbor_02* compared to the baseline.

2) *Scale Drift Analysis*: In monocular VIO, unobservable scale drift is a major source of error, often exacerbated by the poor observability of distant features. We analyzed the Local Scale Factor, defined as the ratio between the estimated and ground-truth translation distances over a sliding window. As shown in Fig. 6, the ORPE-based method significantly alleviates scale drift compared to the baseline. This improvement is primarily attributed to the enhanced gradient fidelity provided by the tangent-space formulation. In the standard reprojection model, linearization errors associated with depth uncertainty often result in biased gradients, causing the estimator to incorrectly adjust the scale and bias terms to minimize residuals. In contrast, ORPE provides more accurate gradient directions by decoupling the measurement geometry from depth, thereby preventing spurious state updates and preserving the consistency of the scale estimate over long trajectories.

3) *Robustness to Feature Depth*: To explicitly verify the advantage of ORPE in handling features at varying distances, we analyzed the average χ^2 error (NIS) of feature updates binned by their triangulated depth. Fig. 7 presents the statistical results. For close-range features (0–5m), both methods perform comparably. However, as the feature depth increases (> 5m), the χ^2 error of the baseline OpenVINS increases dramatically, indicating that the standard visual model struggles to effectively model the uncertainty of distant points. Conversely, OpenVINS

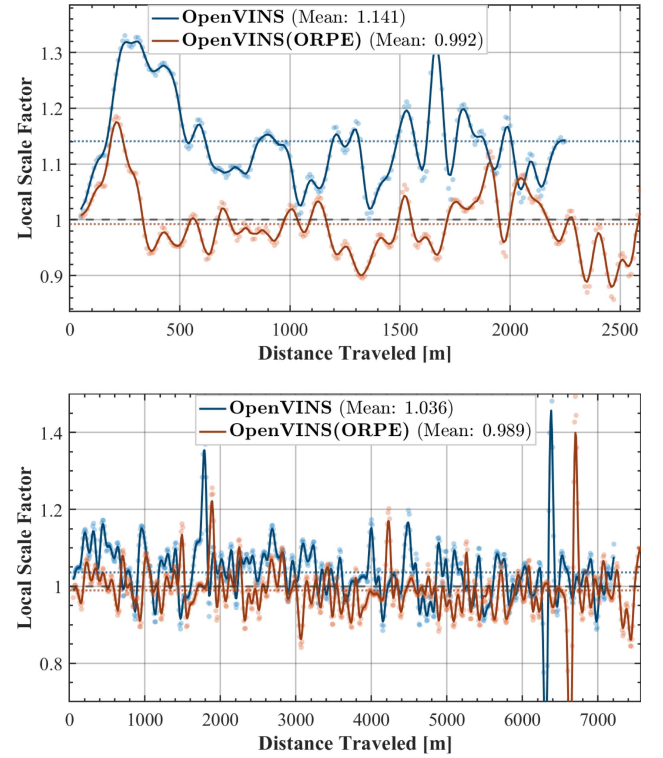


Fig. 6. Evolution of the Local Scale Factor on *neighbor_01* (Up) and *neighbor_02* (Bottom). A value of 1.0 indicates zero scale drift. The standard OpenVINS suffers from significant scale oscillations, whereas OpenVINS (ORPE) effectively constrains the scale factor around the true value, mitigating the drift induced by far-range feature observations.

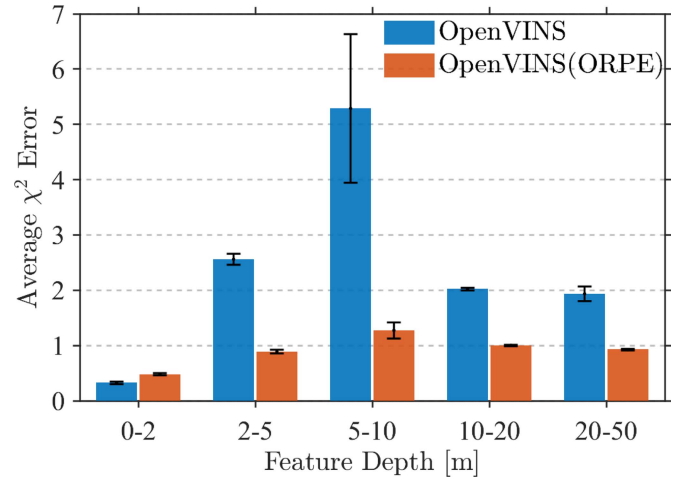


Fig. 7. Average χ^2 error (NIS) of visual measurements with respect to feature depth. The errors are aggregated from the RPNG sequences. While the baseline error grows significantly with feature depth due to linearization issues, the ORPE-based error remains low and invariant to depth changes. This highlights the algorithm's robustness in processing distant environmental features.

(ORPE) maintains a consistently low and stable error level across all depth ranges (up to 50m). This result empirically confirms our theoretical derivation: the Orthogonal Ray Projection minimizes the error in the angular domain, making the residual magnitude invariant to the scalar depth. This property is crucial for large-scale outdoor navigation where distant landmarks are abundant.

E. Computational Efficiency Analysis

Finally, we evaluated the system's runtime performance to verify that the introduction of the ORPE model does not compromise real-time capability. Table III details the average execution time of the key modules in both the MSCKF-based and FGO-based frameworks.

The results indicate that the computational cost of the proposed method is comparable to that of the standard reprojection error model. In the OpenVINS framework, the *Tracking* and *Update* modules show negligible differences in execution time. Although computing the orthogonal basis and the corresponding Jacobian involve additional vector operations, these are lightweight computations. Consequently, the slight increase in the MSCKF update time has no impact on the overall 20-30 Hz frequency of the filter. In the ORB-SLAM3 framework, the frontend *Tracking* thread maintains a similar efficiency to the baseline. Notably, the backend *Local BA* thread even exhibits a slight reduction in runtime, suggesting that the improved residual properties do not add computational complexity to the solver.

VI. CONCLUSION

In this letter, we proposed ORPE to address the numerical instability inherent in standard perspective reprojection. By formulating the residual in the tangent space of the observation ray, ORPE eliminates the nonlinearity induced by perspective division, rendering the measurement model linear with respect to the feature position. We derived the exact analytical Jacobians and integrated the model into both the MSCKF-based OpenVINS and optimization-based ORB-SLAM3 frameworks. Extensive experiments validate that ORPE enhances trajectory accuracy and estimator consistency, particularly in scenarios with weak parallax and large depth uncertainty, while maintaining computational efficiency comparable to standard approaches. However, as a geometric model, its robustness remains constrained by feature tracking quality in dynamic or low-texture scenes. Future work will, therefore, explore integrating ORPE with more robust front-ends or additional sensors to handle such degraded and complex environments.

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